

Multi-Agent Clinical Decision-Making with Indian LLMs

1. Problem Statement

Multi-agent LLM frameworks for clinical decision-making, such as MDAgents, have been developed and evaluated almost exclusively with Western LLM backbones (GPT-4, Claude) and on English-language clinical text. Two gaps follow from this:

- Multi-agent collaboration and escalation strategies for clinical decision-making have not yet been built with Indian LLMs (**e.g., Sarvam, Krutrim**) as the reasoning backbone. Since these models are trained on different data distributions, they may require the collaboration and escalation logic itself to be adapted not just swapped in to work well in a multi-agent debate setting.
- The same frameworks assume clean English input, so they do not address the realistic case of a Hindi or any other regional speaking patient whose query must first be translated, and it is unclear whether adding an explicit translation layer helps or hurts downstream diagnostic accuracy compared to relying on an LLM's native multilingual ability.

Without addressing these gaps, claims about the robustness and generality of multi-agent clinical LLM systems remain untested for Indian-language, India-grounded clinical use cases.

2. Base Research Paper

MDAgents: An Adaptive Collaboration of LLMs for Medical Decision-Making — Kim, Y. et al., NeurIPS 2024. [arXiv:2404.15155](#) | [OpenReview](#) | [github.com/mitmedialab/MDAgents](#)

MDAgents classifies how complex a medical case is, then routes it to a single model, a small panel, or a full committee of LLM agents, mirroring how real clinical teams escalate cases.

Follow-up papers used: TeamMedAgents (structured teamwork extension): [arXiv:2508.08115](https://arxiv.org/abs/2508.08115); Catfish Agent (identifies and fixes agents agreeing too readily) — [arXiv:2505.21503](https://arxiv.org/abs/2505.21503)

3. Novelty: Where This Project Goes Beyond a Backbone Swap

- **Contribution 1:** first MDAgents-style pipeline built with an Indian LLM (Sarvam-30B/105B or Krutrim) as the agent backbone, this has not been done in the literature, which uses only GPT-4/Claude.
- **Contribution 2:** **AI4Bharat IndicTrans2 translation layer** placed before the multi-agent reasoning step, so Hindi language patient input is machine-translated to English before the agents reason

4. Datasets (all public, no ethics approval required)

Dataset	Use	Link
MedQA (USMLE)	Board-exam QA; MDAgents' own primary benchmark, used to reproduce its core result	arXiv:2009.13081
PubMedQA	Biomedical yes/no/maybe QA; MDAgents' second benchmark	pubmedqa.github.io
DDXPlus	Synthetic patient cases with differential diagnoses (CC-BY)	github.com/bruzwen/ddxplus

MedMCQA	194k+ MCQs from Indian medical entrance exams (AIIMS/NEET-PG) — India-grounded content, and the source questions machine-translated to Hindi to test the translation layer	medmcqa.github.io
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4. Architecture

- Case-complexity classifier agent: routes to single-model / panel / full committee (as in MDAgents).
- **Translation layer (new):** Hindi patient input → AI4Bharat IndicTrans2 (Indic-En, open pretrained model) → English text → rest of the pipeline. Built from questions machine-translated English→Hindi via IndicTrans2's En-Indic direction, so no hand-written vernacular data is required.
- Specialist reasoning agents: independently propose an answer on the (translated) English case.
- Aggregator/moderator agent: synthesizes the panel's decision, with a dissent-check step (from the Catfish Agent paper).

5. Evaluation

- Accuracy / Macro-F1 per condition (a/b/c above), per backbone (Indian LLM vs. Western LLM), on all four datasets.
- Single-agent vs. multi-agent delta, and accuracy by case-complexity tier (MDAgents' own routing).
- Agreement-bias rate (Catfish Agent diagnostic) — checked separately for the translated vs. direct-Hindi conditions.
- Translation quality check: compare IndicTrans2's own reported BLEU/chrF on IN22 benchmarks against downstream clinical-accuracy impact, to see if translation-quality loss predicts reasoning errors.

6. Future Work

- **Extend to more Indian languages:** the translation layer currently covers Hindi, but IndicTrans2 supports all 22 scheduled Indian languages; the pipeline can be extended to test other Indian languages beyond Hindi to check whether findings generalize across languages.
- **Extend to voice input:** patient input can be extended beyond text to spoken queries, using AI4Bharat speech models (e.g., IndicWav2Vec/IndicASR) for speech-to-text before the translation and reasoning steps — relevant for low-literacy patients who cannot type.
- **Extend to image input:** the pipeline can be extended to accept image-based clinical inputs (e.g., handwritten prescriptions, lab reports, scanned documents) using AI4Bharat OCR/vision-language models, feeding extracted text into the same translation-and-reasoning pipeline.